

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Use $\tan \theta = \frac{\sin \theta}{\cos \theta}$ correctly (PI) (OE)	AO1.1a	M1	$\tan 15^\circ = \frac{\sqrt{(2-\sqrt{3})}}{\sqrt{(2+\sqrt{3})}}$ $\tan 15^\circ = \frac{2-\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$ $= (2-\sqrt{3})^2$
Squares and multiplies by appropriate <u>conjugate</u> or vice versa	AO1.1b	A1	
Obtains correct values of a and b ($a = 7, b = -4$) If no explicit evidence of <u>conjugate</u> the use of <u>conjugate</u> seen award max 1/3 (M1 A0 R0)	AO2.1	R1	
			$\tan^2 15^\circ = 7 - 4\sqrt{3}$
Total 3 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Compares with $R \cos(x \pm \alpha)$ or $R \sin(x \pm \alpha)$	AO3.1a	M1	$\sqrt{3} \sin x - 3 \cos x = R \sin(x - \alpha)$ $= R \sin x \cos \alpha - R \cos x \sin \alpha$ $R \cos \alpha = \sqrt{3}$ $R \sin \alpha = 3$ $R = \sqrt{12} = 2\sqrt{3}$ $\tan \alpha = \sqrt{3}$ $\alpha = \frac{\pi}{3}$ $y = 2\sqrt{3} \sin\left(x - \frac{\pi}{3}\right) + 4$
	Obtains two correct equations for R and α for example $R \cos \alpha = \sqrt{3}$ $R \sin \alpha = 3$ Must be explicitly seen	AO3.1a	M1	
	Obtains correct R Condone AWR 3.46 PI by description of stretch	AO1.1b	B1	
	Obtains correct α in radians or degrees PI by description of translation	AO1.1b	B1	
	Interprets their values of R	AO3.2a	B1F	

	and α to form an equation of the form $y = R \sin(x \pm \alpha) + 4$ or $y = R \cos(x \pm \alpha) + 4$			Translation $\begin{pmatrix} \frac{\pi}{3} \\ 0 \end{pmatrix}$ Stretch in the y -direction scale factor $2\sqrt{3}$
	Interprets 'their' equation to identify a transformation	AO3.2a	E1F	
	Identifies all required transformations in a correct order CAO	AO3.2a	A1	Translation $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$
(b) (i)	Deduces the least value occurs when their $\sin\left(x - \frac{\pi}{3}\right) = 1$ Using 'their' values of R and α PI by sight of $\frac{1}{2\sqrt{3} + 4}$	AO2.2a	M1	$\frac{1}{\sqrt{3}\sin x - 3\cos x + 4} = \frac{1}{2\sqrt{3}\sin\left(x - \frac{\pi}{3}\right) + 4}$ Least value when $\sin\left(x - \frac{\pi}{3}\right) = 1$ $\therefore$ least value is given by $\frac{1}{2\sqrt{3} + 4} = \frac{2 - \sqrt{3}}{2}$
	Completes rigorous argument to obtain $\frac{1}{2\sqrt{3} + 4}$ and then the given answer	AO2.1	R1	
(ii)	Deduces the greatest value using 'their' values of R and α ACF $\frac{1}{-2\sqrt{3} + 4} = \frac{2 + \sqrt{3}}{2}$	AO2.2a	B1F	Greatest value = $\frac{2 + \sqrt{3}}{2}$
Total 10 marks				

Q3.

Marking Instructions	AO	Marks	Typical Solution
Multiplies numerator and denominator by the conjugate surd of the denominator	AO1.1a	M1	$\frac{(5\sqrt{2} + 2)(3\sqrt{2} - 4)}{(3\sqrt{2} + 4)(3\sqrt{2} - 4)}$
Obtains either numerator or denominator correctly, in expanded or simplified	AO1.1b	A1	$= \frac{30 - 20\sqrt{2} + 6\sqrt{2} - 8}{2}$

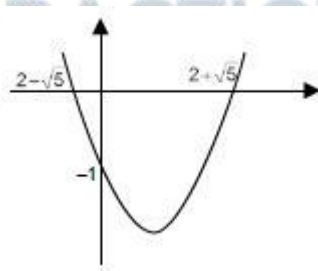
form			$= \frac{22 - 14\sqrt{2}}{2}$
Constructs rigorous mathematical argument to show the required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips NMS = 0	AO2.1	R1	$= 11 - 7\sqrt{2}$
Total 3 marks			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Completes the square twice or applies standard formula	AO1.1a	M1	$(x + 4)^2 + (y - 6)^2 - 16 - 36 = 12$ $(x + 4)^2 + (y - 6)^2 = 64$
	Obtains correct equation	AO1.1b	A1	Centre $(-4, 6)$
	Obtains correct radius and correct coordinates of C Follow through 'their' equation	AO1.1b	A1F	Radius = 8
(b)	Demonstrates a method to find the length OP or OQ (or their squares), or the coordinates P or Q using 'their' values from part (a)	AO3.1a	M1	$OC^2 = 4^2 + 6^2 = 52$ $OP^2 = r^2 - OC^2$ $= 64 - 52 = 12$ $PQ = 2OP$ $2\sqrt{12} = 4\sqrt{3}$
	Uses a circle property that may lead to a solution, eg radius and chord meet at right-angles (evidence for this could be the use of Pythagoras or perpendicular gradients)	AO3.1a	M1	
	Finds OP or OQ or coordinates of P or Q CAO	AO1.1b	A1	
	Obtains length of PQ Follow through from 'their' coordinates of P and Q (Does not need to be in	AO1.1b	A1F	

the required form)			
Sets out a well-constructed mathematical argument, using precise statements and correct use of symbols throughout to show the correct required result in required form Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	
Total 8 marks			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses either of the given coordinates in the given equation (accept product of the roots)	AO1.1a	M1	$k = 4(2 + \sqrt{5}) - (2 + \sqrt{5})^2 = -1$ ALT $k = 4(2 - \sqrt{5}) - (2 - \sqrt{5})^2 = -1$
	Obtains the correct value of k	AO1.1b	A1	ALT $k = (2 - \sqrt{5})(2 + \sqrt{5}) = -1$
(b)	Sketches a graph with the correct shape ✓	AO1.2	B1	
	Deduces correct relative positioning of intersections with axes (must see labels)	AO2.2a	B1	
	Deduces minimum lies to right of y -axis in fourth quadrant	AO2.2a	B1	
				Total 5 marks

Q6.

- (a) (i) $\sqrt{18} = 3\sqrt{2}$
 Condone $k = 3$

B1

(ii) $\frac{2\sqrt{2}}{3\sqrt{2}+4\sqrt{2}}$

attempt to write each term in form $n\sqrt{2}$ with at least 2 terms correct

M1

correct unsimplified

A1

$$= \frac{2}{7}$$

A1

or $\times \frac{\sqrt{2}}{\sqrt{2}}$

M1

integer terms = $\frac{4}{6+8}$

A1

$$= \frac{2}{7}$$

A1

3

(b) $\frac{7\sqrt{2}-\sqrt{3}}{2\sqrt{2}-\sqrt{3}} \times \frac{2\sqrt{2}+\sqrt{3}}{2\sqrt{2}+\sqrt{3}}$

M1

(numerator =) $14 \times 2 - 2\sqrt{6} + 7\sqrt{6} - 3$

correct unsimplified but must simplify

$(\sqrt{2})^2$, $(\sqrt{3})^2$ and $\sqrt{2} \times \sqrt{3}$ correctly

$(\sqrt{2})^2$, $(\sqrt{3})^2$ and $\sqrt{2} \times \sqrt{3}$ correctly

m1

(denominator = $8 - 3$ =) 5

must be seen or identified as denominator

giving $\frac{25+5\sqrt{6}}{5}$

B1

(Answer =) $5 + \sqrt{6}$

$m = 5, n = 6$

A1cso

4

Q7.

(a) $x = 0 \Rightarrow y^2 - 4y - 12 = 0$

sub $x = 0$ & correct quadratic in y

or $(y - 2)^2 = 16$ or $(y - 2)^2 - 16 = 0$

M1

$(y - 6)(y + 2) (= 0)$

correct factors or formula as far as $\frac{4 \pm \sqrt{64}}{2}$
or $y - 2 = \pm \sqrt{16}$

A1

$\Rightarrow y = -2, 6$

condone $(0, -2)$ & $(0, 6)$

A1

3

(b) $(x + 3)^2 - 9 + (y - 2)^2 - 4 (= 12)$

correct sum of square terms **and** attempt to complete squares (or multiply out) PI by next line

M1

$(r^2 =) 9 + 4 + 12$

$(r^2 =) 25$ seen on RHS

A1

$(\Rightarrow r =) 5$

$r = \sqrt{25}$ or $r = \pm 5$ scores A0

A1

3

(c) (i) $(CP^2 =) (2 - - 3)^2 + (5 - 2)^2$

condone one sign slip within one bracket

M1

$\Rightarrow (CP =) \sqrt{34}$
 $n = 34$

A1

2

(ii) $PQ^2 = CP^2 - r^2 = 34 - 25$

Pythagoras used correctly with values FT "their" r and CP

M1

$(\Rightarrow PQ =) 3$

A1

2

[10]

Q8.

(a) (i) $(\sqrt{48} =) 4\sqrt{3}$

condone $n=4$. No ISW.

B1

1

(ii) $\sqrt{12} = 2\sqrt{3}$ and $\sqrt{48} = 4\sqrt{3}$

(FT "their" n) $2x\sqrt{3} = 7\sqrt{3} - 4\sqrt{3}$

M1

$(x =) \frac{7\sqrt{3} - 4\sqrt{3}}{2\sqrt{3}}$

correct quotient unsimplified

or correct equation in integers eg $6x = 21 - 12$

A1

$= \frac{3}{2}$

accept 1.5 but not $\frac{9}{6}$ etc

A1cso

3

Alternative:

$x = \frac{7\sqrt{3} - 4\sqrt{3}}{\sqrt{12}} \times \frac{\sqrt{12}}{\sqrt{12}}$

(M1)

integer terms = $\frac{42 - 24}{12}$

(A1)

$= \frac{3}{2}$

(A1) (3)

(b) $\frac{11\sqrt{3}+2\sqrt{5}}{2\sqrt{3}+\sqrt{5}} \times \frac{2\sqrt{3}-\sqrt{5}}{2\sqrt{3}-\sqrt{5}}$

M1

(numerator =) $22 \times 3 + 4\sqrt{15} - 11\sqrt{15} - 2 \times 5$
correct unsimplified but must simplify
 $(\sqrt{3})^2$, $(\sqrt{5})^2$ and $\sqrt{3} \times \sqrt{5}$ correctly

A1

(denominator = $12 - 5$) 7

must be seen or identified as denominator giving $\frac{56-7\sqrt{15}}{7}$

B1

(Answer =) $8 - \sqrt{15}$
 $m = 8$

A1cso

4

[8]

Q9.

(a) (i) $2(x + 1.5)^2$

OE

M1

$2(x + 1.5)^2 + 0.5$

$2\left(x + \frac{3}{2}\right)^2 + \frac{1}{2}$ OE

A1

2

(ii) (Minimum value is) 0.5

ft their q

B1 ✓

1

(b) (i) $(AB^2 =) (x + 3)^2 + (3x + 9 - 5)^2$

condone one sign error inside one bracket

M1

$$(3x + 4)^2 = 9x^2 + 24x + 16$$

OE

B1

$$AB^2 = x^2 + 6x + 9 + 9x^2 + 24x + 16 = 10x^2 + 30x + 25$$

$$\Rightarrow AB^2 = 5(2x^2 + 6x + 5)$$

AG

A1cso

- (ii) Either $\sqrt{5 \times \text{'their' (a)(ii)}}$ or $5 \times \text{'their' (a)(ii)}$
 using their minimum value from (a)(ii) and 5
 provided "their" (a)(ii) > 0

M1

$$(\text{Minimum length of } AB =) \quad \frac{1}{2}\sqrt{10}$$

A1cso

2

[8]

Q10.

(a) (i) $(3\sqrt{2})^2 = 18$

B1

1

(ii) $(3\sqrt{2} - 1)^2 = \text{'their 18'} - 3\sqrt{2} - 3\sqrt{2} + 1$
 FT their $(3\sqrt{2})^2$

M1

$$= 18 - 3\sqrt{2} - 3\sqrt{2} + 1$$

$$(\quad = 19 - 6\sqrt{2})$$

A1

$$(3 + \sqrt{2})^2 = 9 + 3\sqrt{2} + 3\sqrt{2} + 2$$

$$(\quad = 11 + 6\sqrt{2})$$

B1

$$\Rightarrow \text{Sum} = 30$$

A1cso

4

(b) $\frac{4\sqrt{5} - 7\sqrt{2}}{2\sqrt{5} + \sqrt{2}} \times \frac{2\sqrt{5} - \sqrt{2}}{2\sqrt{5} - \sqrt{2}}$

M1

Numerator = $8(\sqrt{5})^2 - 4\sqrt{5}\sqrt{2} - 14\sqrt{5}\sqrt{2} + 7(\sqrt{2})^2$
correct unsimplified ($= 54 - 18\sqrt{10}$)

M1

Denominator = $(2\sqrt{5})^2 - (\sqrt{2})^2 = 18$
must be seen as denominator

B1

$\Rightarrow \text{Answer} = 3 - \sqrt{10}$

A1cso

4

[9]

Q11.

(a) (i) $\tan \alpha = \frac{4}{3}$

Fraction required
Allow 1.333 (recurring)

B1

1

(ii) 1, 2, $\sqrt{3}$ seen (from Pythagoras)

or

$4 = 1 + \cot^2 \beta$

Use cosec²β = 1 + cot² β

M1

$\tan \beta = -\frac{1}{\sqrt{3}}$

SC B1 $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

A1

Special case B1 for $\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$

2

$$(b) \quad \tan(\alpha + \beta) = \frac{\frac{4}{3} - \frac{1}{\sqrt{3}}}{1 - \frac{4}{3} \left(-\frac{1}{\sqrt{3}} \right)}$$

Use $\tan(\alpha + \beta)$ formula

M1

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ $a \ b \ c \ d$ integers

m1

$$= \frac{4\sqrt{3} - 3}{3\sqrt{3} + 4}$$

$m = 4 \quad n = 3$
or any multiple

A1

3

Alternative $\tan(\alpha + \beta)$

$$\frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\frac{4}{5} \times \left(-\frac{\sqrt{3}}{2} \right) + \frac{3}{5} \times \frac{1}{2}}{\frac{3}{5} \times \left(-\frac{\sqrt{3}}{2} \right) - \frac{4}{5} \times \frac{1}{2}}$$

Use formulae for $\sin(\alpha + \beta)$ and $\cos(\alpha + \beta)$

(M1)

Remove fractions within fractions

Correct manipulation to form $\frac{a+b\sqrt{3}}{c+d\sqrt{3}}$ $a \ b \ c \ d$ integers

(m1)

$$= \frac{-4\sqrt{3} + 3}{-3\sqrt{3} - 4} \quad \left(= \frac{4\sqrt{3} - 3}{3\sqrt{3} + 4} \right)$$

$m = -4 \quad n = -3$
or any multiple

(A1)

(3)

M1 for substituting candidates values for $\tan \alpha$ and $\tan \beta$ into correct formula.
Completely correct or completely correct ft on $\tan \alpha$, $\tan \beta$.

Special case answer is $\frac{12+3\sqrt{3}}{9-4\sqrt{3}}$ or $\times \frac{a}{a}$ where a is integer or $\sqrt{3}$ for M1m1A0.

[6]

Q12.

$$\frac{5\sqrt{3}-6}{2\sqrt{3}+3} \times \frac{2\sqrt{3}-3}{2\sqrt{3}-3}$$

M1

(Numerator =) $30 - 15\sqrt{3} - 12\sqrt{3} + 18$
correct $(= 48 - 27\sqrt{3})$

m1

(Denominator = $12 - 9$) 3
must be seen as denominator

B1

$$\left(\frac{48 - 27\sqrt{3}}{3} \right) = 16 - 9\sqrt{3}$$

CSO; accept $16 + - 9\sqrt{3}$

A1

[4]

Q13.

(a) 27

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B1

1

(b) $\frac{4\sqrt{3}+3\sqrt{7}}{3\sqrt{3}+\sqrt{7}} \times \frac{3\sqrt{3}-\sqrt{7}}{3\sqrt{3}-\sqrt{7}}$

M1

(Numerator =) $36 + 9\sqrt{21} - 4\sqrt{21} - 21$

expanding numerator
condone one slip or omission

m1

(Denominator =) 20
must be seen as denominator

B1

$$\frac{15 + 5\sqrt{21}}{20} = \frac{3 + \sqrt{21}}{4}$$

$$m = 3, n = 4 \quad \text{condone} \quad \frac{3}{4} + \frac{\sqrt{21}}{4}$$

A1cso

4

[5]

Q14.

(a) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

Condone numerator as $\tan x + \tan x$

B1

$$2 \tan x + \tan x (1 - \tan^2 x) = 0$$

Multiplying throughout by their denominator

M1

$$\tan x = 0$$

$$\text{or } (2 + 1 - \tan^2 x) = 0 \Rightarrow \tan^2 x = 3$$

AG Must show $\tan x = 0$ and $\tan^2 x = 3$

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A1

3

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$$\tan 2x = \frac{\sin 2x}{\cos 2x} = \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} + \frac{\sin x}{\cos x} = 0$$

(B1)

$$2 \sin x \cos^2 x + \sin x (\cos^2 x - \sin^2 x) = 0$$

$$\sin x (2 \cos^2 x + \cos^2 x - \sin^2 x) = 0$$

(M1)

$$\left. \begin{aligned} \Rightarrow \sin x &= 0 \\ \Rightarrow \tan x &= 0 \end{aligned} \right\} \text{ and } \left. \begin{aligned} 3 \cos^2 x &= \sin^2 x \\ \tan^2 x &= 3 \end{aligned} \right\}$$

(A1)

(3)

(ii) $x = 60$ **AND** $x = 120$

*Condone extra answers outside interval
eg 0 and 180*

B1

1

(b) (i) $2 \sin x \cos x = \cos x f(x)$

Where $f(x) = \cos^2 x - \sin^2 x$

or $2 \cos^2 x - 1$ or $1 - 2 \sin^2 x$

M1

$$2 \sin x \cos x = \cos x (1 - 2 \sin^2 x)$$

A1

$(\cos x \neq 0) \quad 2 \sin x = 1 - 2 \sin^2 x$

$$2 \sin^2 x + 2 \sin x - 1 = 0$$

AG

A1

3

(ii) $\sin x = \frac{-2 \pm \sqrt{4 - 4 \times 2 \times (-1)}}{2 \times 2}$

*Correct use of quadratic formula or completing
the square or correct factors*

M1

$$\sin x = \frac{-2 \pm 2\sqrt{3}}{4}$$

$\sqrt{12}$ *must be simplified and must have $\pm$.*

A1

$$\left. \begin{aligned} \sin x &= \frac{-1 - \sqrt{3}}{2} \text{ has no solution} \\ \sin x &= \frac{\sqrt{3} - 1}{2} \end{aligned} \right\}$$

Reject one solution and state correct solution.

E1

3

[10]

Q15.

(a) (i) $\sqrt{48} = 4\sqrt{3}$

condone $k = 4$ stated

B1

1

(ii) $\frac{4\sqrt{3} + 6\sqrt{3}}{2\sqrt{3}}$

*attempt to write each term in form $k\sqrt{3}$ with
at least 2 terms correctly obtained*

M1

correct unsimplified in terms of $\sqrt{3}$ only.

A1

$= 5$

must simplify fraction to 5

A1cso

3

Alternative 1 $\times \frac{\sqrt{12}}{\sqrt{12}} \left(\text{or } \times \frac{\sqrt{3}}{\sqrt{3}} \right)$ M1

correct with integer terms $= \frac{24 + 36}{12}$ A1

$= 5$ A1cso

Alternative 2 $\frac{\sqrt{48} + \sqrt{108}}{\sqrt{12}}$ M1

$= \sqrt{4} + \sqrt{9}$ A1

$= 5$ A1cso

Alternative 3 $\sqrt{\frac{48}{12}} + 2\sqrt{\frac{27}{12}}$ M1

$= 2 + 2\sqrt{\frac{9}{4}}$ A1

$= 5$ A1cso

*if hybrid of methods used, award M1
 and most appropriate first A1
 NMS (answer =) 5 scores full marks*

(b) $\frac{1-5\sqrt{5}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}$

M1

(numerator =) $3 - \sqrt{5} - 15\sqrt{5} + 25$
correct unsimplified but must write
 $5\sqrt{5}\sqrt{5} = 25$ PI by 28 seen later

m1

(denominator = $9 - 5 =$) 4
must be seen as denominator

B1

giving $\frac{28-16\sqrt{5}}{4}$

(answer =) $7 - 4\sqrt{5}$
 $m = 7, n = -4$

A1

4

[8]

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Q16.

(a) $\sqrt{50} = 5\sqrt{2}; \sqrt{18} = 3\sqrt{2}; \sqrt{8} = 2\sqrt{2}$

At least two of these correct

or $\times \frac{\sqrt{8}}{\sqrt{8}}$ or $\left(\times \frac{\sqrt{2}}{\sqrt{2}} \right)$ or $\sqrt{\frac{25}{4}} + \sqrt{\frac{9}{4}}$

M1

$\frac{5\sqrt{2} + 3\sqrt{2}}{2\sqrt{2}}$

*any **correct** expression all in terms of $\sqrt{2}$
 or with denominator of 8, 4 or 2*

simplifying numerator eg $\frac{\sqrt{400} + \sqrt{144}}{8}$

A1

Answer = 4

CSO

A1

3

(b) $\frac{2\sqrt{7} - 1}{2\sqrt{7} + 5} \times \frac{2\sqrt{7} - 5}{2\sqrt{7} - 5}$
OE

M1

numerator = $4 \times 7 - 2\sqrt{7} - 10\sqrt{7} + 5$

expanding numerator
(condone one error or omission)

m1

denominator = 3

(seen as denominator)

B1

Answer = $11 - 4\sqrt{7}$

A1

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[7]

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Q17.

(a) $(3 - \sqrt{5})^2 = 9 - 6\sqrt{5} + (\sqrt{5})^2$

Allow one slip in one of these terms
M0 if middle term is omitted

M1

= $14 - 6\sqrt{5}$

A1

2

(b) $\frac{(3 - \sqrt{5})^2}{1 + \sqrt{5}} \times \frac{1 - \sqrt{5}}{1 - \sqrt{5}}$

$$\text{or } \dots \times \frac{\sqrt{5}-1}{\sqrt{5}-1}$$

M1

$$14 + 6\sqrt{5}\sqrt{5} - 6\sqrt{5} - 14\sqrt{5}$$

$$(\text{= } 44 - 20\sqrt{5})$$

*Expanding **their** numerator*
(condone one error or omission)

m1

(Denominator) = -4
Must be seen as denominator

B1

$$(\text{Answer}) = -11 + 5\sqrt{5}$$

Accept "answer = $5\sqrt{5} - 11$ "

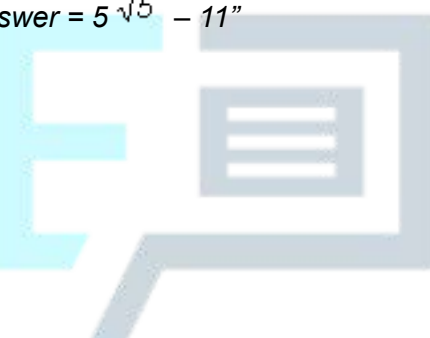
A1

4

[6]

Q18.

$$(a) \quad \frac{7+\sqrt{5}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}$$


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Multiply by $\frac{3-\sqrt{5}}{3-\sqrt{5}}$ or $\frac{\sqrt{5}-3}{\sqrt{5}-3}$

M1

$$\text{Numerator} = 21 + 3\sqrt{5} - 7\sqrt{5} - (\sqrt{5})^2$$

Condone one slip $16 - 4\sqrt{5}$

m1

Denominator = $9 - 5 = 4$
(Or $5 - 9 = -4$ from other conjugate)

B1

$$\text{Answer} = 4 - \sqrt{5}$$

CSO

A1

4

(b) $\sqrt{45} = 3\sqrt{5}$

B1

$$\frac{20}{\sqrt{5}} = \frac{20\sqrt{5}}{5}$$

*May score if combined as one expression
Must have 5 in denominator*

M1

Sum = $7\sqrt{5}$

A1

3

[7]

Q19.

(a) $\frac{5 + \sqrt{7}}{3 - \sqrt{7}} \times \frac{3 + \sqrt{7}}{3 + \sqrt{7}}$

M1

Numerator = $15 + 5\sqrt{7} + 3\sqrt{7} + 7$

Condone one error or omission

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m1

Denominator = $9 - 7 (= 2)$

Must be seen as the denominator

B1

(Answer =) $11 + 4\sqrt{7}$

A1

4

(b) $(2\sqrt{5})^2 = 20$ or $(3\sqrt{2})^2 = 18$

Either correct

B1

their $(2\sqrt{5})^2 - (3\sqrt{2})^2$

Condone missing brackets and x^2

M1

$$(x^2 = 20 - 18)$$

$$x^2 = 2 \Rightarrow B1, M1$$

$$(\Rightarrow x = 2)\sqrt{2}$$

$$\pm \sqrt{2} \text{ scores A0}$$

Answer only of 2 scores B0, M0

Answer only of $\sqrt{2}$ scores 3 marks

A1

3

[7]

Q20.

(a) $5\sqrt{8} = 10\sqrt{2}$

Or $\frac{5\sqrt{16} + 6}{\sqrt{2}}$ gets B1

B1

$$\frac{6}{\sqrt{2}} = \frac{6\sqrt{2}}{2} (= 3\sqrt{2})$$

then M1 for rationalising; and A1 answer

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M1

Answer = $13\sqrt{2}$ 2025 Exam Papers Practice. All rights reserved.

$$n = 13$$

A1

3

(b) $\frac{\sqrt{2} + 2}{3\sqrt{2} - 4} \times \frac{3\sqrt{2} + 4}{3\sqrt{2} + 4}$

Multiplying top & bottom by $\pm (3\sqrt{2} + 4)$

M1

$$\text{Numerator} = 6 + 6\sqrt{2} + 4\sqrt{2} + 8$$

Multiplying out (condone one slip)

m1

Denominator = $18 - 16 (= 2)$

B1

Final answer = $5\sqrt{2} + 7$

A1

4

[7]

Q21.

(a) $xy = 6$

B0 for $\sqrt{36}$ or ± 6

B1

1

(b) $\frac{y}{x} = \frac{2\sqrt{3}}{\sqrt{3}}$ or $\sqrt{\frac{12}{3}}$ or $\sqrt{\frac{4}{1}}$ or $\frac{\sqrt{12}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

Allow M1 for ± 2

M1

$= 2$

A1

2

(c) $x^2 + 2xy + y^2$ or $(\sqrt{3} + 2\sqrt{3})^2$ correct

or $(\sqrt{3} + \sqrt{12})(\sqrt{3} + \sqrt{12})$ expanded as 4 terms – no more than one slip

M1

Correct with 2 of x^2 , y^2 , $2xy$ simplified

Correct but unsimplified – one more step

A1

$3 + 2\sqrt{36} + 12$ or $3^2 \times 3$ or $(3\sqrt{3})^2$
 $= 27$

A1

3

[6]

Q22.

(a) $\frac{\sqrt{5}+3}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2}$

Multiplying top & bottom by $\pm(\sqrt{5}+2)$

M1

Numerator = $5+3\sqrt{5}+2\sqrt{5}+6$

Multiplying out (condone one slip) $\pm(\sqrt{5+3})(\sqrt{5+2})$

M1

$=5\sqrt{5}+11$

A1

Final answer $=5\sqrt{5}+11$

With clear evidence that denominator = 1

A1

4

(b) (i) $\sqrt{45} = 3\sqrt{5}$

B1

1

(ii) $\sqrt{20} = \sqrt{4}\sqrt{5}$ or $4\sqrt{5} = \sqrt{4} \times \sqrt{20}$

Both sides

M1

or attempt to have equation with $\sqrt{5}$ or $\sqrt{20}$ only

$[x \cdot 2\sqrt{5} = 7\sqrt{5} - 3\sqrt{5}]$ or $x \cdot \sqrt{20} = 2\sqrt{20}$

or $x = \sqrt{4}$

A1

$x = 2$

A1

3

[8]

Q23.

(a) $\frac{\sqrt{63}}{3} = \sqrt{7}$ or $\frac{3\sqrt{7}}{3}$

or $\frac{(\sqrt{7}\sqrt{63} + 14 \times 3)}{3\sqrt{7}}$

B1

$\frac{14}{\sqrt{7}} = 2\sqrt{7}$ or $\frac{14\sqrt{7}}{7}$

or $\frac{\sqrt{7}}{\sqrt{7}} ()$ M1

B1

$\Rightarrow \text{sum} = 3\sqrt{7}$

$\Rightarrow$ correct answer with all working correct A2

B1

3

(b) Multiply by $\frac{\sqrt{7} + 2}{\sqrt{7} + 2}$

M1

Denominator = $7 - 4 = 3$

A1

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Numerator = $(\sqrt{7})^2 + \sqrt{7} + 2\sqrt{7} + 2$

multiplied out (allow one slip) $9 + 3\sqrt{7}$

m1

Answer = $\sqrt{7} + 3$

A1

4

[7]

Q24.

(a) $(\sqrt{5})^2 + 2\sqrt{5} - 2\sqrt{5} - 4 = 1$

M1

2

*Multiplying out or difference of two squares attempted
Full marks for correct answer / no working*

A1

(b) $\sqrt{8} = 2\sqrt{2} \quad ; \quad \sqrt{18} = 3\sqrt{2}$

Either correct

M1

Answer = $5\sqrt{2}$

Full marks for correct answer / no working

A1

2

[4]

Q25.

(a) $4(\sqrt{5})^2 + 12\sqrt{5} - \sqrt{5} - 3$
Multipled out

M1

$4(\sqrt{5})^2 = 4 \times 5 \quad (= 20)$

At least 3 terms with $\sqrt{5}$ term

B1

Answer = $17 + 11\sqrt{5}$

A1

3

(b) Either $\sqrt{75} = \sqrt{25}\sqrt{3}$ or $\sqrt{27} = \sqrt{9}\sqrt{3}$

Or multiplying top and bottom by $\sqrt{3}$

M1

Expression = $\frac{5\sqrt{3} - 3\sqrt{3}}{\sqrt{3}}$
or $\frac{\sqrt{255} - \sqrt{81}}{3}$ or $\sqrt{25} - \sqrt{9}$ or $5 - 3$

A1

$$= 2$$

CSO

A1

3

[6]

Q26.

- (a) $(\sqrt{12})^2 - 2^2$ attempt to multiply out

May have $\sqrt{12}$ terms

M1

$$= 12 - 4 = 8$$

A1

2

- (b) $2\sqrt{3}$

B1

1

- (c) Multiplying top and bottom by $\sqrt{12} + 2$

Or $\sqrt{3} + 1$ etc

B1

$$\text{Numerator} = 12 + 4\sqrt{12} + 4$$

*At least 3 terms multiplied out on top
OE in*

M1

$$\text{Expression} = \frac{16 + 4\sqrt{12}}{8} \quad \text{or} \quad \frac{16 + 8\sqrt{3}}{8}$$

*ft denominator from (a); or correct
but numerator correct (unsimplified)*

A1ft

$$= 2 + \sqrt{3}$$

A1

4

[7]

Q27.

(a) Midpoint $\left(\frac{6+2}{2}, \frac{5-1}{2}\right) = (4, 2)$

One coordinate correct unsimplified

M1

Both correct and simplified

A1

2

(b) $AB^2 = (6-2)^2 + (5+1)^2$

Pythagoras used (condone one slip)

M1

$= (16 + 36) = 52$

52 or $\sqrt{52}$ seen

A1

$\Rightarrow AB = 2\sqrt{13}$

A1

3

(c) (i) Gradient $AB = (5 - -1) / (6 - 2)$

Must be y on top and subtraction (6 - 2)

M1

$= \frac{6}{4} = \frac{3}{2} = 1.5$

Any correct equivalent

A1

2

(ii) $y - 5 = m(x - 6)$ or $y + 1 = m(x - 2)$

or $y = mx + c$ and attempt to find c

M1

$2y - 10 = 3x - 18$ etc leading to

$3x - 2y = 8$

AG (be convinced)

A1

2

- (d) Attempt to eliminate x or y

$$7x = 28 \text{ etc}$$

M1

$$x = 4$$

A1

$$y = 2$$

C is point $(4, 2)$

A1

3

[12]

Q28.

- (a) Multiplied out

$$3 + 1 + 2\sqrt{3}$$

At least 3 terms with $\sqrt{3}$ term

M1

$$= 4 + 2\sqrt{3}$$

$$m = 4, n = 2$$

A1

2

- (b) Multiplying top and bottom by $\sqrt{3} + 1$

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M1

$$\text{Denominator} = 3 - 1 = 2$$

B1

$$\text{Expression} = \frac{4 + 2\sqrt{3}}{2}$$

$$= 2 + \sqrt{3}$$

$$\text{CSO } m = 2, n = 1$$

A1

3

[5]

Q29.

(a) $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

Full method

M1

A1 if four terms correct or just one slip

A2,1

3

(b) (i) $(1 + \sqrt{5})^4 = 1 + 4\sqrt{5} + 6(\sqrt{5})^2 + 4(\sqrt{5})^3 + (\sqrt{5})^4$

Substitute. $\sqrt{5}$ for x .

M1

$$= 1 + 4\sqrt{5} + 6(5) + 4(5\sqrt{5}) + (25)$$

Two of 3 terms shown in brackets

A1ft

$$= 56 + 24\sqrt{5}$$

AG *Be convinced*

A1CSO

3

(ii) $\log_2 (1 + \sqrt{5})^4 = \log_2 [8(7 + 3\sqrt{5})]$

M1

$$= \log_2 8 + \log_2 (7 + 3\sqrt{5})$$

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$$= 3 + \log_2 (7 + 3\sqrt{5})$$

SC B1 *Change to base 10 and verify as per model marked script*

A1CSO

3

[9]